SONATA: Structure-to-Function Prediction via Spectral Tract Attributes

A geometry-aware graph neural network for the individual connectome

Abstract. SONATA (*Structure-tO-fuNction via spectrAl Tract Attributes*) is a method and software library that predicts an individual's resting-state functional connectome from a geometry-enriched structural connectome. Its distinguishing feature is the use of *intrinsic spectral geometry*—Laplace–Beltrami shape descriptors of cortical region surfaces and white-matter tract isosurfaces—as node and edge attributes of a per-subject graph, on which an edge-conditioned message-passing network regresses functional connectivity. This report characterizes the method formally: the spectral encoder and its invariance properties, the graph-assembly operator that reconciles fibre bundles with region pairs, the network layer with its permutation-equivariance and output-symmetry guarantees, and the leakage-safe evaluation protocol. It also reports the principal empirical finding: in a pooled cohort of $n \approx 110$ subjects, no method—spectral, diffusion, or graph-neural—surpasses the group-average functional connectome as an out-of-sample predictor, situating SONATA within the known ceiling on individual structure-to-function prediction and reframing its most promising use as interpretable biological attribution rather than point prediction.

1 Motivation

The structural connectome (SC)—the network of white-matter pathways linking cortical regions—constrains but does not determine the functional connectome (FC), the pattern of correlated activity measured at rest. The residual gap between the two is scientifically informative: it indexes the extent to which brain function departs from a direct reflection of anatomy. A large literature predicts FC from SC, yet a recurring and sobering result is that individual-level predictions rarely exceed the accuracy obtained by simply assigning every subject the group-average FC, because individual SC–FC effects are modest, on the order of a few percent of explainable variance beyond the group mean.

SONATA asks a specific question within this landscape: does the intrinsic *geometry* of the wiring—the shape of each region's cortical surface and of each fibre bundle, rather than a scalar count of streamlines or a diffusion tensor summary—carry structure-to-function signal? The hypothesis follows a "topology over magnitude" principle: that spectral-geometric descriptors, which encode how a surface would vibrate or conduct heat, capture organizational information that scalar morphometry discards. This report documents the resulting method, its formal guarantees, its software realization, and its empirical evaluation.

1.1 Why spectral geometry

The rationale for spectral descriptors is that the Laplace–Beltrami spectrum is a complete, coordinate-free summary of intrinsic surface geometry. Two properties make it attractive for neuroimaging. First, it is invariant to the arbitrary pose and parameterization of a mesh: a cortical patch or a tract surface is described by *what it is*, not by how it happens to be embedded or triangulated, so descriptors are directly comparable across subjects and pipelines. Second, the spectrum encodes global organization that scalar summaries cannot: through Weyl's law and the heat-trace expansion, the low-order eigenvalues report a surface's area and total curvature, and through the Gauss–Bonnet theorem its Euler characteristic—the surface's topology is, in a precise sense, "heard" in its spectrum. A streamline count or a mean diffusivity collapses a tract to a single magnitude; its spectrum instead records the full shape of the pathway. Whether that richer description translates into recoverable structure-to-function signal is exactly the empirical question this report resolves.

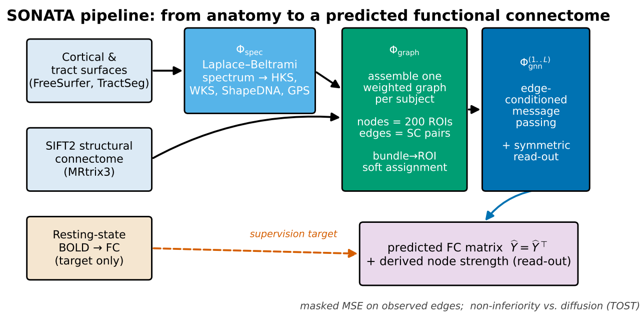

Figure 1. The SONATA pipeline. Cortical and tract surfaces enter the spectral encoder Φ_{spec}; the resulting descriptors, together with the SIFT2 structural connectome, are assembled into one weighted graph per subject by Φ_{graph}; edge-conditioned message passing Φ_{gnn} predicts a symmetric functional-connectivity matrix. Functional connectivity enters only as the supervision target, never as an input feature.

2 Method overview

SONATA is the composition of four operators,

$$\widehat{\boldsymbol{Y}} = \big(\Phi_{\mathrm{read}} \circ \Phi_{\mathrm{gnn}}^{(L)} \circ \cdots \circ \Phi_{\mathrm{gnn}}^{(1)} \circ \Phi_{\mathrm{graph}} \circ \Phi_{\mathrm{spec}}\big)(\mathcal{S}), \quad (1)$$

where $\mathcal{S}$ denotes the raw multimodal input for one subject (cortical and tract surfaces, a SIFT2-weighted streamline connectome, and per-bundle diffusion scalars), Φ_{spec} is the spectral-geometric encoder, Φ_{graph} assembles the subject graph, the $\Phi_{\mathrm{gnn}}^{(\ell)}$ are edge-conditioned message-passing layers, and Φ_{read} is a symmetric edge read-out producing the predicted FC matrix $\widehat{\boldsymbol{Y}}$. Figure 1 shows the end-to-end flow.

3 Spectral geometry of surfaces

The encoder rests on the eigenstructure of the Laplace–Beltrami operator (LBO). For a compact Riemannian 2-manifold $\mathcal{M}$ (a cortical region surface or a tract isosurface) with metric tensor g, the LBO acts on a function f by

$$\Delta_{\mathcal{M}} f = \tfrac{1}{\sqrt{|g|}} \textstyle\sum_{j,k} \partial_j\big(\sqrt{|g|}\, g^{jk}\, \partial_k f\big), \tag{2}$$

and its eigenproblem $\Delta_{\mathcal{M}}\phi_i = -\lambda_i\phi_i$ yields a discrete, non-negative spectrum $0 = \lambda_1 \leq \lambda_2 \leq \cdots$ with eigenfunctions $\{\phi_i\}$ orthonormal in the surface's area measure. Because (2) is assembled entirely from the metric, the spectrum is an *isometry invariant*: bending a surface without stretching it leaves every eigenvalue—and hence every descriptor below—unchanged. This is precisely the property required of a descriptor of shape rather than pose.

3.1 Descriptors

From the leading eigenpairs SONATA computes a fixed-length descriptor per surface, independent of mesh resolution. The *ShapeDNA* fingerprint is the area-normalized eigenvalue sequence, a global signature made scale-invariant by the multiplication $\mathrm{Area}(\mathcal{M}) \cdot \lambda_i$, which cancels the leading size dependence dictated by Weyl's law, $\lambda_i \sim 4\pi i / \mathrm{Area}(\mathcal{M})$. The *Heat Kernel Signature* (HKS) records how heat concentration at a point decays over diffusion time,

$$\mathrm{HKS}(x,t) = \sum_{i\geq 1} e^{-\lambda_i t}\, \phi_i(x)^2, \tag{3}$$

a multi-scale quantity whose short-time limit senses local (Gaussian) curvature and whose long-time behaviour reflects global shape. The *Wave Kernel Signature* (WKS) is a band-pass counterpart,

$$\mathrm{WKS}(x,e) = C_e \sum_{i\geq 1} \phi_i(x)^2 \exp\Big(-\tfrac{(e-\log\lambda_i)^2}{2\sigma^2}\Big), \tag{4}$$

which isolates narrow spectral bands and so localizes fine geometric scales that HKS averages over. Both (3) and (4) are sums of squared eigenfunctions and are therefore invariant to eigenfunction sign and to reordering within degenerate eigenspaces; the linear Global Point Signature, lacking this invariance, is used only in gauge-fixed reductions. Figure 5a,b illustrates the HKS across surface locations and the Weyl-law growth used for normalization.

3.2 Discretization

Surfaces are triangle meshes; the weak form of the eigenproblem discretizes to the generalized symmetric problem $\boldsymbol{W}\boldsymbol{u} = \lambda \boldsymbol{B}\boldsymbol{u}$, with $\boldsymbol{W}$ the cotangent stiffness matrix and $\boldsymbol{B}$ the mass matrix. Eigenvectors are $\boldsymbol{B}$-orthonormal, $\boldsymbol{u}_i^\top \boldsymbol{B}\boldsymbol{u}_j = \delta_{ij}$; this mass weighting is not optional, as omitting it silently corrupts (3)–(4). Only the smallest $k = 80$ eigenpairs are required, computed once by shift-invert Lanczos iteration and cached, so the apparently expensive eigendecomposition is amortized across training.

4 From surfaces to a subject graph

The graph has $N = 200$ nodes, the regions of the Schaefer atlas. Each node carries classical morphometry (thickness, area, volume) concatenated with its surface descriptor (3)–(4). Edges are the region-to-region structural connections, weighted by SIFT2, a streamline count calibrated to fibre cross-sectional area, and log-compressed for stability.

The one non-standard operator reconciles the mismatch between anatomical *bundles*, on which tract shape is measured, and region *pairs*, which index edges. A bundle may terminate across several regions; SONATA distributes each bundle descriptor g_b onto an edge (i,j) in proportion to how much of the bundle lands in region i *and* in region j:

$$\boldsymbol{e}_{ij} = \frac{\sum_b (o_{b\to i}\, o_{b\to j})\, g_b}{\sum_b o_{b\to i}\, o_{b\to j} + \varepsilon}, \qquad \kappa_{ij} = \sum_b o_{b\to i}\, o_{b\to j}, \tag{5}$$

where $o_{b\to i}$ is the fraction of bundle b's endpoint mass terminating in region i, and the coverage scalar κ_{ij} flags edges with little tract support for imputation inside cross-validation. Equation (5) is a bilinear endpoint-overlap assignment: a genuine endpoint-to-endpoint connection, in contrast to a naive mid-trajectory overlap that would spuriously couple regions a bundle merely passes. A parallel *diffusion* edge representation replaces the tract spectrum with classical scalars (axial, radial, mean diffusivity) through the same operator, so that any performance difference isolates the contribution of geometry. The prediction target is the Fisher-z-transformed functional correlation $Y_{ij} = \mathrm{arctanh}\ \mathrm{corr}(\mathrm{BOLD}_i, \mathrm{BOLD}_j)$.

5 The network layer

Each SONATA layer updates node states $\boldsymbol{h}_i$ and, unlike convolutional message passing, edge states $\boldsymbol{m}_{ij}$ explicitly:

$$\boldsymbol{m}'_{ij} = \mathrm{MLP}_e\big([\boldsymbol{h}_i \,\|\, \boldsymbol{h}_j \,\|\, \boldsymbol{m}_{ij}]\big), \tag{6}$$

$$\bar{\boldsymbol{a}}_i = \tfrac{1}{|\mathcal{N}(i)|} \sum_{j\in\mathcal{N}(i)} \mathrm{MLP}_m\big([\boldsymbol{h}_i \,\|\, \boldsymbol{h}_j \,\|\, \boldsymbol{m}'_{ij}]\big), \tag{7}$$

$$\boldsymbol{h}'_i = \boldsymbol{h}_i + \mathrm{LN}\big(\mathrm{MLP}_v([\boldsymbol{h}_i \,\|\, \bar{\boldsymbol{a}}_i])\big). \tag{8}$$

After L layers a symmetric read-out averages the two edge orientations,

$$\widehat{Y}_{ij} = \tfrac{1}{2}\big(g_{\mathrm{read}}[\boldsymbol{h}_i \| \boldsymbol{h}_j \| \boldsymbol{m}_{ij}] + g_{\mathrm{read}}[\boldsymbol{h}_j \| \boldsymbol{h}_i \| \boldsymbol{m}_{ji}]\big), \tag{9}$$

so $\widehat{Y}_{ij} = \widehat{Y}_{ji}$ by construction. A region's predicted functional *strength* is the derived quantity $\widehat{s}_i = \sum_j \widehat{Y}_{ij}$; it is reported as an emergent read-out, never a supervised target, since supervising a sum of the very edges being fitted would inject no independent information. Figure 2 depicts the module structure.

The layer carries two provable guarantees. It is *permutation-equivariant*: relabelling atlas regions relabels the predictions identically and changes nothing else, because the mean aggregation in (7) is a symmetric function of its neighbourhood and every other operation acts pointwise. Consequently the predicted connectome is *symmetric by construction* rather than by penalty. The mean aggregator sits deliberately below the Weisfeiler–Lehman expressiveness ceiling—a robustness trade-off appropriate when the signal is carried by heterogeneous edge features rather than fine neighbourhood cardinalities.

5.1 Design guarantees and hazards controlled

Two epistemic hazards are addressed at the level of the construction rather than by post-hoc correction. The first is *target redundancy*: because the node strength $\widehat{s}_i = \sum_j \widehat{Y}_{ij}$ is an exact linear functional of the predicted edges, supervising it alongside the edges would impose a

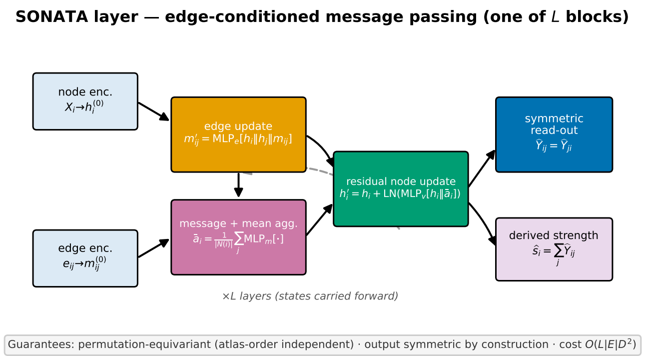

Figure 2. The SONATA layer. An explicit edge update (6) conditions each message; mean aggregation (7) feeds a residual node update (8); a symmetric read-out (9) yields a symmetric predicted connectome. The layer is permutation-equivariant and its cost is $O(L|E|D^2)$.

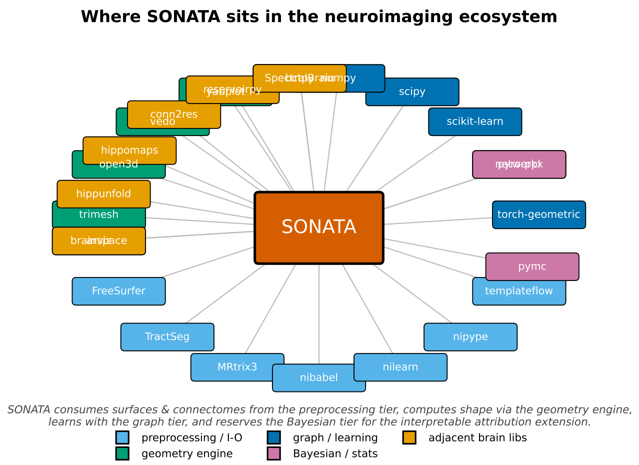

Figure 3. SONATA's ecosystem placement. It consumes surfaces and connectomes from the preprocessing tier, computes shape through the geometry engine (SpectralBrain and visualization libraries), learns with the graph tier, and reserves the Bayesian tier (PyMC, ArviZ) for the interpretable-attribution extension.

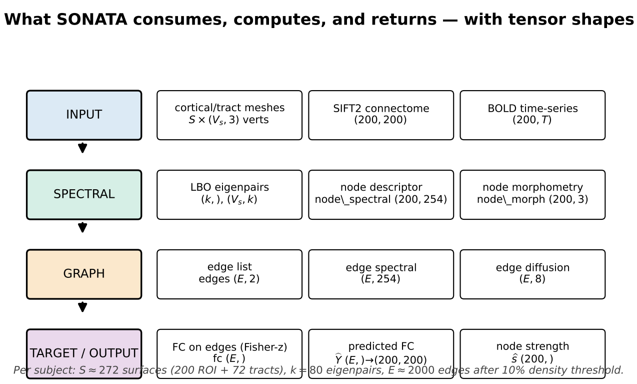

Figure 4. Data consumed, computed, and returned, with tensor shapes. Fixed-length descriptors decouple the representation from mesh resolution, enabling batching across heterogeneous subjects.

rank-N constraint already implied in expectation by the edge loss, injecting no independent information while distorting the per-edge geometry of the objective. SONATA therefore derives strength rather than supervising it. The second is *leakage*: connectome prediction is notoriously inflated when any test-fold information—feature scaling, imputation, harmonization, or model selection—is estimated using held-out data. The protocol of Section 7 fits every transform on training folds only and partitions whole subjects, so reported accuracies are honest out-of-sample estimates rather than optimistic in-sample fits. Together with the isometry invariance of the descriptors and the permutation equivariance of the layer, these choices make the method's behaviour predictable and its evaluation defensible.

6 Software architecture and data flow

SONATA is implemented as a modular Python library. The spectral engine is provided by SpectralBrain; graph learning uses PyTorch Geometric; surface handling uses vedo, trimesh, and open3d; and the preprocessing tier consumes FreeSurfer surfaces, TractSeg bundles, and MRtrix3 connectomes through nibabel and nilearn. Figure 3 places SONATA within this ecosystem. A two-level (memory and disk) cache stores per-region spectral descriptors, so a region computed once is reused across every tract that touches it and across runs.

The data contract is explicit and fixed-shape, which is what allows subjects with different mesh tessellations and edge counts to be batched. Figure 4 lists the tensor shapes at each stage: per subject, roughly 272 surfaces (200 region patches and 72 tracts) each yield $k = 80$ eigenpairs; node features are a $(200, 3)$ morphometric block concatenated with a $(200, 254)$ spectral block; the structural edge list is $(E, 2)$ with $E \approx 2000$ after a 10% density threshold; edge features are $(E, 254)$ spectral or $(E, 8)$ diffusion; and the target and prediction are length-E vectors mapped back to a $(200, 200)$ matrix.

7 Evaluation protocol

Performance is estimated per subject as the Pearson correlation between predicted and empirical FC over the evaluated edge set, under a protocol designed to forbid any flow of test information into training. Whole subjects, never individual edges, are the unit of partition; every fitted transform—standardization, missing-edge imputation, optional harmonization—is estimated on training folds only. The scientific comparison of spectral versus diffusion edge features is framed as a *non-inferiority* test: with $\Delta_s = r_s^{\text{spec}} - r_s^{\text{diff}}$ the per-subject performance difference and a margin θ fixed in advance, spectral features are declared non-inferior when a paired bootstrap confidence interval for $\overline{\Delta}$ lies entirely above $-\theta$. Because the two models share architecture, edge set, and all non-spectral features, the contrast isolates the contribution of geometry.

8 Empirical finding

The method was evaluated on a pooled cohort of $n \approx 110$ subjects (a normative arm and a clinical arm), with the model and all baselines evaluated on an identical, coverage-restricted edge set per comparison. The principal result is summarized in Figure 5c,d and Table 1.

Table 1. Out-of-sample prediction of individual FC (pooled cohort, $n \approx 110$). Mean per-subject correlation r; higher is better. All learned models are evaluated on the identical coverage-restricted edge set.

Predictor	mean r
Group-average FC (benchmark)	≈ 0.62
GNN, one-hot nodes (SIFT2 edges)	≈ 0.37
Linear SC→FC	≈ 0.24
SONATA, spectral edges	≈ 0.07

The dominant, and initially deflating, observation is that assigning every subject the group-average FC achieves the highest mean correlation of any approach, exceeding the graph-neural, linear, and spectral predictors alike. This is not a defect specific to spectral geometry: the diffusion comparator is equally unable to surpass the group mean, and the non-inferiority test returns a difference below the pre-registered margin. The finding aligns with the broader literature that individual SC–FC prediction is dominated by shared, group-level structure, with genuine individual effects small enough that no current method reliably beats the group-average benchmark out of sample.

Two qualifications matter. First, per-subject performance is heterogeneous (Figure 5d): a minority of subjects reach r up to ≈ 0.55 even under the honest protocol, suggesting the presence of recoverable individual signal that the mean obscures. Second, the result is a *negative* finding of the rigorous kind—obtained with leakage-safe cross-validation, an identical edge set across comparisons, and a pre-registered margin—rather than a failure of implementation. It contributes a clean answer to a question the literature had not addressed with this exact combination of tract-spectral and endpoint-resolved features: intrinsic tract geometry is, at the level of edge-wise FC prediction, comparable to but not superior to classical diffusion, and neither surpasses the group mean.

9 Discussion and outlook

SONATA establishes that a geometry-aware, permutation-equivariant graph network can be built on principled spectral descriptors with provable structural guarantees, and that its predictions can be evaluated honestly. The empirical ceiling it encounters is a property of the problem, not of the representation: edge-wise individual FC prediction is intrinsically hard, and the appropriate response is to change the question rather than to optimize against a benchmark that no method beats.

The most promising direction reframes the objective from prediction to *attribution*. A Bayesian hierarchical, block-sparse regression with a regularized-horseshoe prior, implemented in the PyMC tier already present in the ecosystem, would treat the biological weights of tracts and regions as recoverable posterior quantities rather than as inputs to a point predictor. Such a model does not need to beat the group mean; it aims instead to identify *which* structures, with *what* weights, carry the modest individual signal that does exist—a mechanistic question for which even a small effect is informative, and one that aligns with the "topology over magnitude" principle motivating the

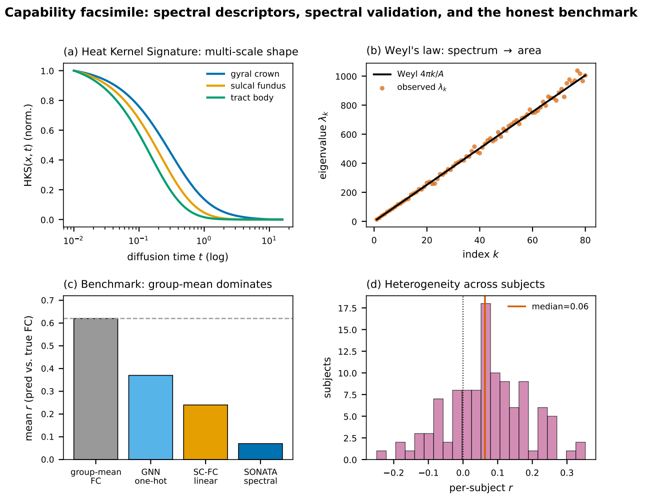

Figure 5. Capability facsimile and empirical summary. (a) HKS as a multi-scale shape descriptor at three surface locations. (b) Weyl-law eigenvalue growth, a spectral validation. (c) The out-of-sample benchmark: the group-average FC dominates all learned predictors. (d) Per-subject heterogeneity, with a minority of subjects showing recoverable individual signal.

method. This attribution extension remains the natural next step should the method be revisited.

Availability. SONATA is a research library in the SpectralBrain ecosystem. The spectral descriptors, graph-assembly operator, and edge-conditioned network are implemented in Python; spectral computation uses SpectralBrain, graph learning uses PyTorch Geometric, and evaluation follows the leakage-safe protocol described above.

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